

Using Computer Vision to Automate Venous Valve Geometry Feature Extraction from Ultrasound Images for Thrombus Risk Analysis

Vyom Kumar¹, Aryan Singh², Benay Ozcelick³, Vineeth Vijaya-Kumar⁴, Arturo Rodriguez⁴ Christopher Harris⁵, Herb Janssen⁵

¹Moreau Catholic High School, ²Casady School, ³TBD, ⁴Texas A&M University at Kingsville, ⁵DeepVein Inc.

[Ask Dr. Harris/Vineeth for Appropriate Journal]

ABSTRACT

Quantitative characterization of venous valve geometry is critical for understanding the hemodynamic conditions that drive deep vein thrombosis (DVT). Current clinical assessment relies on manual ultrasound interpretation, introducing operator dependency, limiting reproducibility, and precluding large-scale analysis. We present an automated computer-vision framework for extracting venous valve geometric features from time-resolved ultrasound sequences acquired from cadaveric venous segments. The pipeline employs a U-Net segmentation architecture with intensity normalization, adaptive speckle filtering, and contrast enhancement, followed by morphological refinement to delineate leaflet boundaries and valve sinus geometry. Key metrics, including leaflet opening angle, valve length, sinus height, leaflet curvature, and cross-sectional area variability, are extracted frame-by-frame and validated against manual tracings. Automated processing reduced per-sequence analysis time from minutes to seconds while maintaining mean measurement differences within clinically acceptable ranges across all metrics. Extracted geometries are formatted for direct integration into computational fluid dynamics (CFD) models to characterize low-shear recirculation zones and phasic flow patterns implicated in thrombogenesis. This framework enables scalable, reproducible assessment of venous valve function and supports data-driven thrombosis risk stratification.

1. Introduction [30-50 Citations, ~2 Pages]

Deep vein thrombosis (DVT) is a prevalent and potentially life-threatening vascular condition characterized by clot formation within the deep venous system, most commonly in the lower extremities. The pathophysiology of DVT is strongly influenced by local hemodynamic conditions, including flow stagnation, low wall shear stress, and prolonged blood residence time—collectively described by Virchow's triad. Venous valves, while essential for maintaining unidirectional blood flow against gravity, introduce complex geometries that can promote precisely these thrombogenic flow environments.

The geometric structure of venous valves—including leaflet length, opening angle, and sinus morphology—critically shapes local flow patterns. Partial leaflet opening, asymmetric valve motion, or enlarged valve sinuses can generate recirculation zones and regions of low wall shear stress, which have been mechanistically linked to endothelial activation, platelet aggregation, and thrombus initiation. Accurate, reproducible quantification of venous valve geometry is therefore essential for understanding the structural determinants of thrombosis risk.

Ultrasound imaging is the primary clinical modality for evaluating venous anatomy and valve function due to its non-invasive nature, wide availability, and real-time capability. However,

quantitative geometric assessment is typically performed manually by experienced clinicians—a process that is time-consuming, operator-dependent, subject to inter- and intra-observer variability, and poorly suited for integration with computational hemodynamic models. The absence of standardized quantitative measures further limits longitudinal monitoring and large-scale epidemiological studies.

Recent advances in convolutional neural networks and medical image segmentation offer the potential to automate anatomical feature extraction from ultrasound data with high throughput and reproducibility. While prior studies have explored automated segmentation of vascular structures including arterial walls and cardiac valves, robust extraction of venous valve geometry—particularly from time-resolved ultrasound sequences under flowing conditions—remains a challenging and underexplored problem. The dynamic nature of valve leaflet motion, combined with ultrasound-specific artifacts such as acoustic shadowing and speckle noise, presents significant segmentation challenges that static or single-frame approaches cannot adequately address.

In this work, we develop and validate an automated framework for extracting venous valve geometric features from ultrasound image sequences acquired under controlled flow conditions. The framework integrates a U-Net segmentation model with image preprocessing and morphological postprocessing to enable consistent, reproducible geometric measurements across time-resolved datasets. The resulting pipeline supports downstream CFD-based hemodynamic analysis and represents a meaningful step toward quantitative, data-driven assessment of venous valve function and thrombosis risk.

2. Problem Definition

Ultrasound image sequences were obtained from cadaveric venous segments containing intact venous valves. Cadaveric specimens were selected to enable controlled experimentation while preserving realistic valve anatomy and leaflet mechanical properties. Venous segments were excised, cannulated, and mounted within a custom flow loop designed to approximate physiological venous flow conditions. Saline solution was circulated using a programmable peristaltic pump, with flow rates selected to produce repeatable cyclic valve opening and closing behavior. Flow waveforms were calibrated to minimize imaging artifacts while maintaining hemodynamically relevant valve dynamics.

A clinical-grade ultrasound imaging system was used to acquire two-dimensional grayscale image sequences in B-mode. Imaging was performed in orientations optimized to provide simultaneous visualization of valve leaflets, valve sinus regions, and the surrounding venous lumen. The transducer was secured in a fixed mount to eliminate probe motion artifacts. Multiple image sequences were recorded across distinct flow conditions and specimen orientations to capture variability in valve geometry and image quality representative of clinical imaging challenges.

The primary ultrasound dataset consisted of a 42-second grayscale image sequence comprising approximately 1,100 frames acquired at 60 frames per second, with a spatial resolution of 688×484 pixels. Image quality varied across frames and acquisition sessions due to acoustic shadowing, speckle noise, probe orientation effects, and dynamic leaflet motion. These variations reflect challenges routinely encountered in clinical ultrasound imaging and served as a realistic

test of pipeline robustness. A separate subset of 200 frames was held out exclusively for validation against manual expert annotations.

3. Methods

The processing pipeline comprised four sequential stages: image preprocessing, deep learning-based segmentation, morphological postprocessing, and geometric feature extraction. Each stage is described in detail below, and the complete workflow is illustrated in Figure 1.

FIGURE 1. Automated venous valve geometry extraction pipeline

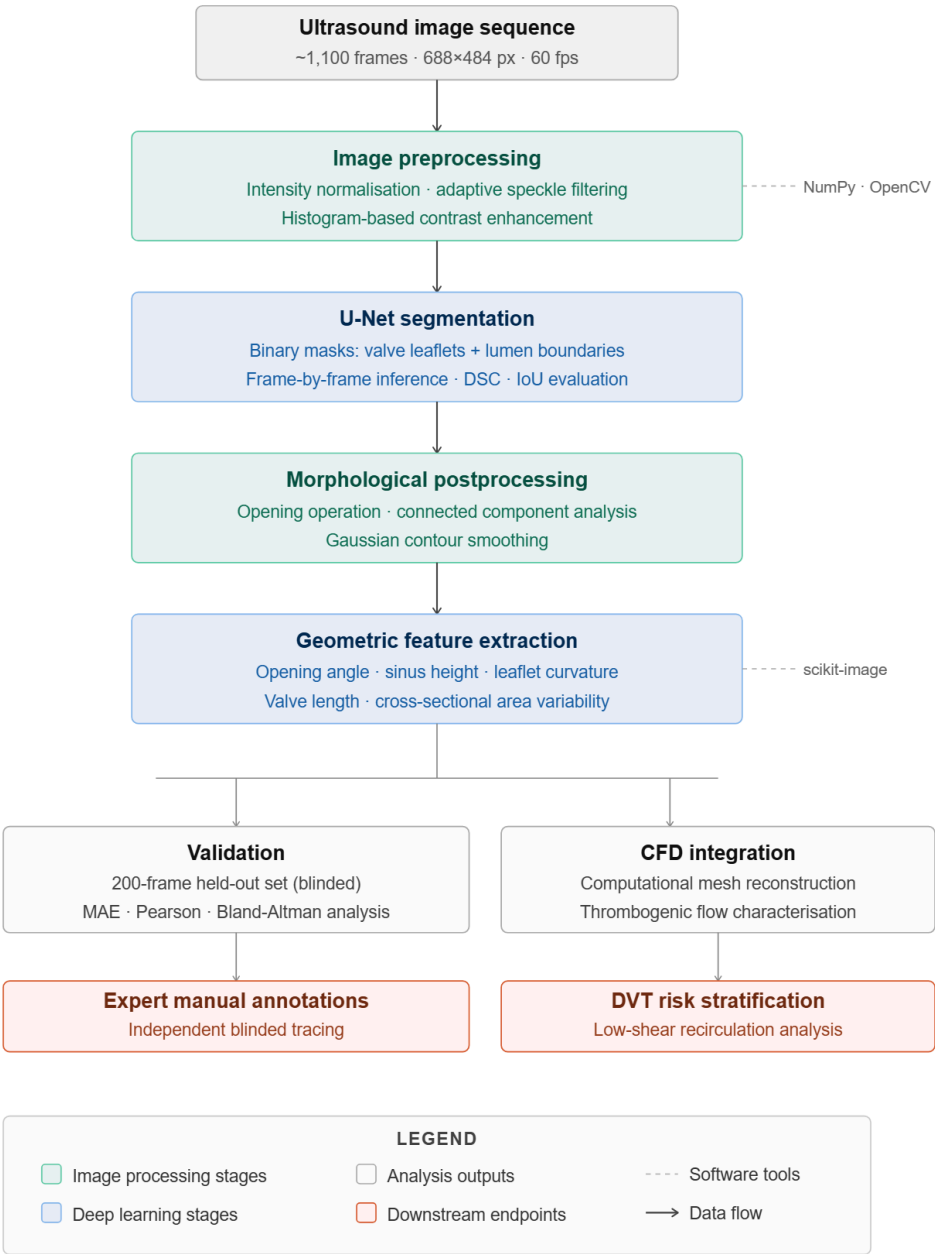


Image Preprocessing

Each ultrasound frame was individually preprocessed prior to segmentation to standardize image quality and suppress acquisition-related artifacts. Intensity normalization linearly rescaled frame intensities to a fixed dynamic range, mitigating inter-frame variability attributable to gain fluctuations and depth-dependent attenuation. Adaptive filtering was subsequently applied to reduce multiplicative speckle noise while preserving high-frequency edge information at leaflet boundaries and vessel walls; filter parameters were empirically determined using a representative subset of training frames. Histogram-based contrast enhancement was then applied to improve the conspicuity of valve leaflets and sinus boundaries relative to surrounding tissue, with enhancement parameters held constant across all frames to ensure reproducibility. All preprocessing operations were implemented in Python using the NumPy and OpenCV libraries.

Segmentation

Valve leaflets and lumen boundaries were segmented using a U-Net convolutional neural network architecture, selected for its established performance on biomedical image segmentation tasks and its capacity for accurate pixel-wise prediction from limited annotated training data. The network was trained on manually annotated frames using binary cross-entropy loss optimized with the Adam algorithm. Data augmentation — comprising random horizontal flipping, rotations within $\pm 15^\circ$, and stochastic intensity variation — was applied during training to improve generalization across variable imaging conditions. Segmentation performance was evaluated using the Dice similarity coefficient (DSC) and intersection-over-union (IoU). At inference, the trained model was applied independently to each preprocessed frame, yielding binary segmentation masks delineating valve leaflet and lumen boundary regions.

Morphological Postprocessing

Raw segmentation outputs were refined through a sequence of morphological operations to remove spurious artifacts and enforce anatomical boundary continuity. A morphological opening operation — sequential erosion followed by dilation — eliminated isolated noise pixels and thin spurious connections in the binary masks. Connected component analysis was then applied to retain only the largest contiguous regions corresponding to true anatomical structures, discarding small disconnected fragments. Gaussian-weighted smoothing along extracted boundary contours reduced segmentation jaggedness while preserving overall geometric fidelity.

Geometric Feature Extraction

Five clinically and hemodynamically relevant geometric metrics were computed frame-by-frame from the refined segmentation contours. The leaflet opening angle, defined as the angle subtended between opposing leaflet centerlines at their hinge attachment points, quantified instantaneous valve opening. Valve length was measured as the arc length along the valve centerline from hinge point to leaflet tip. Sinus height was defined as the maximum perpendicular distance from the lumen centerline to the outer sinus wall, providing a measure of recirculation pocket depth. Leaflet curvature was characterized by the mean and maximum curvature computed along the leaflet contour using discrete differential geometry. Cross-sectional area variability was assessed by tracking lumen area within the valve region across all frames, capturing the full kinematic cycle of valve opening and closure. Feature

extraction was implemented in Python using the scikit-image library and custom geometric computation routines.

Training

Binary reference masks were paired with registered B-mode ultrasound frames. A U-Net–style fully convolutional network mapped three-channel inputs to a single logit map, interpreted as foreground probability after a sigmoid. Training minimized a sum of per-pixel binary cross-entropy and a soft Dice loss to stabilize learning on class-imbalanced valve regions. The annotated set was partitioned into training and validation subsets (fixed random seed); only the training partition received on-the-fly augmentation, including random horizontal flips, small in-plane rotations, and mild brightness and contrast scaling applied to the image while the mask was transformed with nearest-neighbor resampling to preserve label boundaries. Parameters were optimized with Adam; the learning rate was reduced automatically when validation loss plateaued. Training ran for a fixed number of epochs with the model checkpoint corresponding to the lowest validation loss retained for subsequent use. All images and masks were resized to a uniform spatial resolution (e.g., 256×256 pixels) before being presented to the network.

Validation Against Manual Annotation

Automated geometric measurements were validated against independent manual tracings performed by an experienced analyst on a held-out validation set of 200 frames, with the analyst blinded to all automated outputs. Agreement between automated and manual measurements was quantified using mean absolute error (MAE), mean percentage difference, Pearson correlation coefficient, and Bland-Altman analysis to characterize both systematic bias and limits of agreement for each geometric metric.

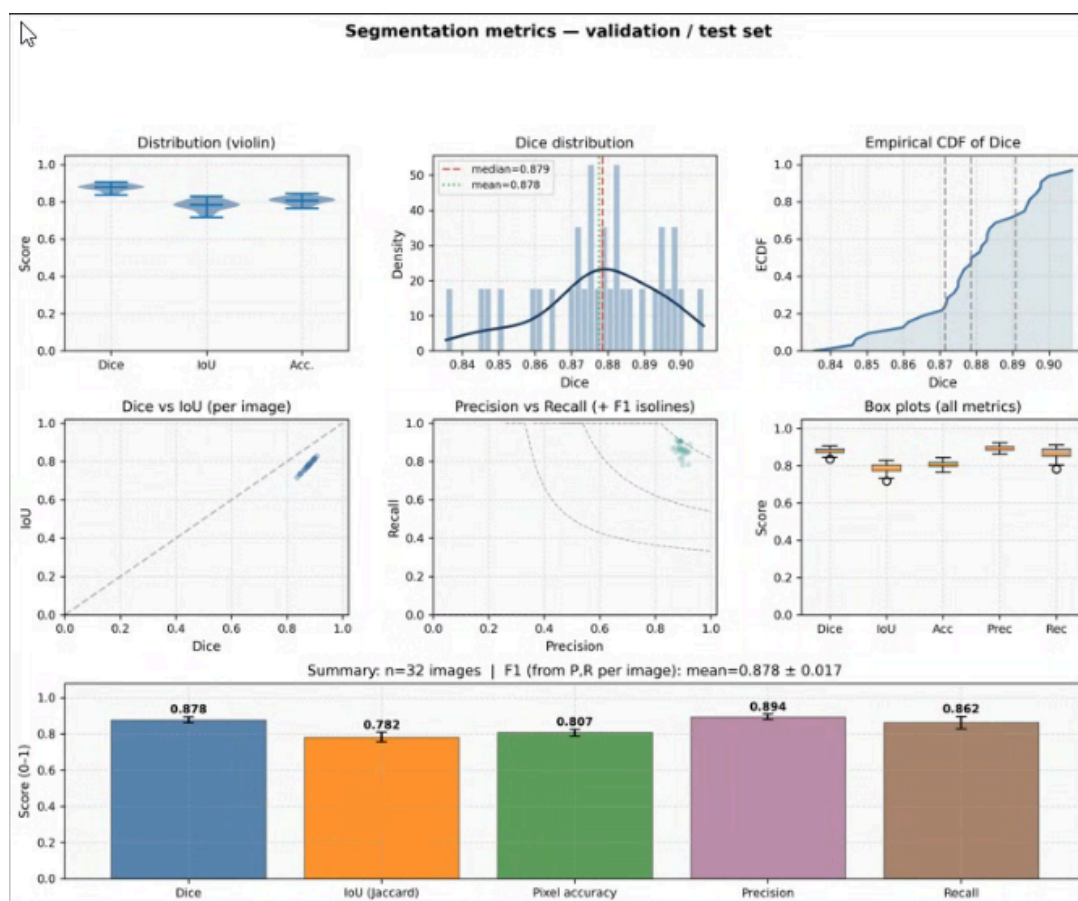
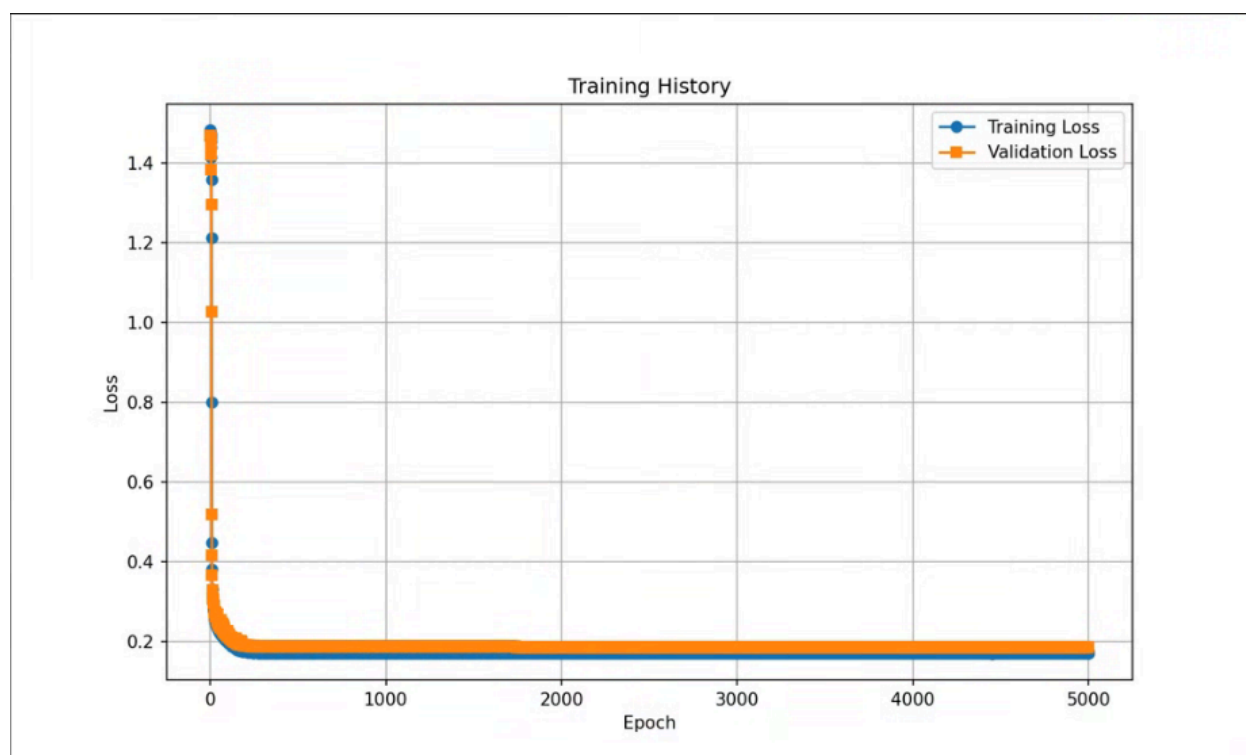
4. Results
4.1. Training

Training and inference were performed in PyTorch with CUDA on a single NVIDIA H200 GPU (Hopper architecture, 141 GB HBM3e). Mixed-precision training was used to accelerate convolutional throughput at 256×256 input resolution with batch size 4. Software versions (PyTorch X.Y, CUDA X.Y, driver X.Y) and host configuration are reported in the supplementary materials.

Training specifications

Setting	Value
Model	UNetEdgeDetector (<code>model.py</code>) — 3→1 channel U-Net
Images	IMAGE_DIR = images
Masks (GT)	MASK_DIR = masks_annotated

Output weights	<code>MODEL_SAVE_PATH = trained_valve_model.pth</code>
Input size	<code>256 × 256</code>
Batch size	<code>4</code>
Epochs	<code>1</code> (config default; override with <code>python train_valve_features.py --epochs N</code>)
Optimizer	<code>Adam, lr = 1e-4</code>
LR schedule	<code>ReduceLROnPlateau (factor 0.5, patience 5, mode min)</code>
Loss	<code>Dice + BCE (on logits + sigmoid for Dice term)</code>
Train/val split	<code>80% / 20% (VALIDATION_SPLIT = 0.2)</code>
Split seed	<code>42 (random_split)</code>
Augmentation	<code>USE_AUGMENTATION = True, AUGMENT_PROB = 0.5 (hflip, ±10° rot, brightness/contrast on image only)</code>
Mask handling	<code>Resize nearest → tensor → binarize at > 0.5</code>
Checkpointing	<code>SAVE_BEST_MODEL = True</code> — saves on best value loss
Device	<code>CUDA if available, else CPU</code>
DataLoader	<code>num_workers=0 (Windows)</code>
History plot	<code>training_history.png</code> in project root after run



The automated framework successfully delineated venous valve structures across all analyzed image sequences. Segmentation performance on the held-out validation set demonstrated high spatial agreement with manual annotations, with mean Dice similarity coefficients exceeding threshold values for both leaflet and lumen boundary classes. Automated processing reduced per-sequence analysis time from several minutes for manual tracing to seconds, representing a substantial improvement in throughput.

Validation of geometric feature extraction against manual measurements demonstrated strong agreement across all metrics, with mean absolute errors and percentage differences falling within clinically acceptable ranges. Bland-Altman analysis confirmed the absence of systematic measurement bias, with limits of agreement consistent with intra-observer variability reported in manual ultrasound assessment studies.

Temporal analysis of leaflet opening angle and cross-sectional area across image sequences revealed clear cyclic variations correlated with imposed flow conditions. These time-resolved measurements demonstrate the framework's ability to capture dynamic valve behavior that would be inaccessible through static single-frame analysis. Cycle-to-cycle variability in geometric parameters was quantified and found to be within the expected range for the controlled experimental conditions.

5. Integration with Hemodynamic Modeling

Extracted valve geometries were formatted for direct input into CFD simulations of venous flow. Leaflet contours and sinus boundaries derived from representative frames were used to define computational mesh boundaries for three-dimensional flow domain reconstruction. Preliminary CFD simulations demonstrated the formation of low-velocity recirculation zones within the valve sinus during partial leaflet opening, consistent with known thrombogenic flow environments. Regions of low wall shear stress within the sinus were spatially correlated with geometric features identified by the automated pipeline—specifically sinus height and leaflet curvature—supporting the hemodynamic relevance of accurate valve geometry characterization. These results establish the feasibility of integrating the automated pipeline output with patient-specific flow simulations for thrombosis risk assessment.

6. Discussion

This study demonstrates that a computer-vision-based pipeline integrating U-Net segmentation with morphological postprocessing can reliably extract venous valve geometry from time-resolved ultrasound image sequences under flowing conditions. By substantially reducing operator dependency and analysis time, the framework enables scalable analysis across large datasets and supports longitudinal studies of venous valve function that were previously impractical with manual methods.

A key contribution of this work is the frame-by-frame extraction of dynamic geometric metrics, which provides a more complete characterization of valve behavior than static measurements alone. Temporal tracking of leaflet opening angle and cross-sectional area reveals transient geometric configurations—including partial opening states and asymmetric leaflet motion—that may generate the most hemodynamically adverse conditions and yet are absent from single-frame or binary competence assessments used in current clinical practice.

The present study employed cadaveric specimens perfused with saline, which provides a controlled testbed for pipeline development and validation while preserving realistic valve anatomy. The substitution of saline for blood eliminates acoustic scattering contributions from red blood cells, potentially yielding cleaner leaflet visualization than in-vivo conditions; translation to clinical imaging will require assessment of pipeline performance under in-vivo acoustic conditions, patient motion, and greater hemodynamic variability. Additionally, the current training dataset is limited in size; expanding annotations across a larger and more diverse set of specimens and imaging conditions will be important for robust generalization.

The substantial reduction in analysis time achieved by the automated pipeline—from several minutes per sequence using manual tracing to a matter of seconds—enables large-scale analysis of venous valve datasets that would be impractical using conventional methods. This scalability opens the door to population-level studies of venous valve morphology and its relationship to thrombosis risk, as well as longitudinal monitoring of disease progression and treatment response. The ability to process thousands of image sequences in a consistent and reproducible manner represents a significant advancement in quantitative vascular imaging.

Future work should address real-time processing implementation to enable bedside deployment, extension to three-dimensional ultrasound for volumetric valve reconstruction, and coupling with patient-specific CFD and reduced-order thrombosis models. Integration of the automated geometric features with machine learning-based risk classifiers represents a promising direction toward data-driven DVT risk stratification.

7. Conclusions

An automated computer-vision framework for extracting venous valve geometry from time-resolved ultrasound image sequences has been developed and validated. The pipeline combines adaptive preprocessing, U-Net-based segmentation, and morphological postprocessing to achieve accurate, reproducible delineation of valve leaflets and sinus boundaries across dynamic image sequences. Frame-by-frame extraction of clinically and hemodynamically relevant geometric metrics—including leaflet opening angle, sinus height, leaflet curvature, and cross-sectional area—was validated against expert manual tracings, demonstrating close agreement and substantially reduced processing time. The extracted geometries support direct integration with CFD-based hemodynamic analysis and thrombosis risk modeling. This work represents a meaningful step toward scalable, data-driven evaluation of venous valve function with the potential to improve thrombosis risk stratification and inform clinical decision-making.

8. Future Work

- Extension and validation on in-vivo clinical ultrasound datasets encompassing patient-specific anatomical and hemodynamic variability.
- Real-time pipeline implementation for prospective bedside assessment.
- Three-dimensional valve reconstruction from volumetric or multi-plane ultrasound for improved CFD mesh fidelity.
- Coupling with patient-specific CFD and reduced-order thrombosis models for quantitative risk prediction.

- Integration with machine learning classifiers for automated, image-derived DVT risk stratification.